Grade 10 LS end of year MEMO

Question 6

6.1.1. Mitochondria

6.1.2. Nucleus

6.1.3. Phospholipid bilayer or cell membrane

6.2.1. Carbon fixation

6.2.2. Decomposers are responsible for breaking down dead or decaying matter.

6.2.3. In the carbon cycle plants photosynthesise taking in carbon and releasing oxygen. In the water cycle plants uptake water in order to photosynthesise.

6.2.4. Sunlight is important as it allows plants to 'fix' carbon in the photosynthetic process.

6.2.5. Nutrient cycles are important as they are responsible for maintaining homeostasis within the environment. Certain elements need to be broken down to be used by other organisms and this process will occur in various nutrient cycles.

6.2.6. Yes they do. Humans play a role in all the cycles as we respire releasing CO2, we burn fossil fuels and factories release carbon into the atmosphere. We play a role in the water cycle as we have a direct impact on the condition of water sources.

Question 7

7.1. Interphase

7.2. It is important as the cell needs to duplicate itself in mitosis so it is imperative to have the same DNA and organelles throughout both cells. It is also vital in order to check for any repairs that may need to take place.

7.3. The cell will abort the duplication.

7.4. Chromosomes must be in the centre of the cell, there must be four chromosomes and the drawing should be neat.

Question 8

8.1. Plants are found in various habitats and what is useful to one plant may not be beneficial to another. For example, a plant in the desert would not have big leaves as that may increase water loss. Hairs and thorns may help for protection as well as reduce water loss.

8.2. Any adaption with a relevant explanation (thorns, size, hairs, colour).

8.3. The phloem is responsible for carrying food throughout the plant whilst the xylem is responsible for carrying water.

Question 9

9.1. Any two producers (must be plants) and any one primary producer.

9.2. There may be conflict as these two organisms are sharing a common living space and ultimately resources. This is due to the fact that the study showed elephants were more likely to prefer areas outside of the protected areas due to the food availability.

9.3. Any four organisms (must be forest animals and it must be a food WEB).

9.4. An example is the Cape Floristic Region. It means that it is an area that has a high abundance of different organisms, some of which are endemic to that area meaning they are only found there and nowhere else in the world hence the term ‘global’.

9.5. Fragmented means separated. A fragmented landscape means that there are gaps between the landscape (this could be farms or other land that has been deforested).

9.6.1. Scientists would have gathered this data via a population census. They would have counted the number of people living in a specific place then they would have possibly counted the number of elephants or evidence of elephants in an area.

9.6.2. The independent variable would be the area the scientists chose and the dependant variable would be the number of people or elephants present.

9.6.3. The table needs to include a heading √√, Column and row headings need to be correct √√, data represented needs to be correct √√, humans and elephants need to be represented √√, table must be neat √√

Question 10

10.1. A = active site

10.2. The diagram represents a lock and key model. This means that the substrate fits perfectly in to the enzyme. Each enzyme therefore only targets a certain substrate. √√

10.3. Enzymes in our bodies are catalysts. A catalyst will speed up a reaction by helping to lower the activation energy needed to start a reaction. Each enzyme molecule has a special place called the active site where another molecule, called the substrate, fits. The lower the activation energy for a reaction, the faster the rate.

10.4. At a higher temperature the enzyme may no longer operate or may be denatured therefore, decreasing the rate of reaction.

Question 11

11.1. Heading √√, Correct number of sugar units √√√√√, all plants shown √, units on axis √, axis titles √√.

11.2. Plant 4.

11.3. Water = 16, CO_2 = 40, Light = 30

Sugar = 3

11.4. It needs 60 more units of light.

11.5. The light.

11.6. Photosynthesis is biologically important as it helps convert CO_2 to O_2 and it also creates glucose for a plant to survive.

Question 12

12. Digestion has been spoken about including chemical and physical √√√√

Sources have been consulted and referenced in the writing √√√

Importance of nutrients have been discussed √√√

There is a logical flow √√

Scientific terminology has been used √

Own knowledge has been used √√